

DD_SALES-01 - Lead, Territory, and Franchise Sales Management

Version: v1.0

Status: Draft for review

Bundle: 10_other

Depends on:

- DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0
- FOUNDATION_AUTH
- FOUNDATION_KAFKA
- FOUNDATION_FILE_STORAGE
- DD_EM-02-Contractor_and_Staff_Registry-v1.0
- DD_EM-02-Contractor_Team_Agent_Registry-v1.0
- DD_ILM-CFG-01-Customer_Service-v1.0
- DD_ILM-CFG-02-Tech_Region_Registry-v1.0
- DD_RLM-CFG-01-HomePass_Configuration-v1.0
- DD_SIP-01-Package_Management-v1.0
- DD_FUL-02-Order_Capture-v1.0
- DD_NOT-01-Notification_Service-v1.3

1. Purpose

This DD defines the commercial sales module used by the Outdoor Sales / Franchise App.

SALES-01 owns commercial leads, lead assignment, sales territories, franchise sales teams, sales activity tracking, and lead-to-order conversion references. It prevents the sales frontend from storing leads locally or misusing FUL-02 as a lead CRM.

FUL-02 remains the owner of submitted customer orders. SALES-01 owns the pre-order commercial pipeline.

2. Scope

Included:

Capability	Purpose
Sales territory	Define commercial areas, route areas, and franchise sales coverage.
Lead capture	Capture prospects before they become customers/orders.
Lead assignment	Assign leads to sales agents, teams, or franchise supervisors.
Lead qualification	Track interest, serviceability, package interest, consent, and duplicate risk.
Sales activity	Capture visit/call/contact attempts and outcomes.
Lead conversion	Convert qualified lead into FUL-02 order request.
Sales attribution	Preserve agent/franchise/channel attribution on the resulting order.
Commission event basis	Emit immutable sales attribution events; actual payout calculation may be deferred.

Out of scope:

Out of scope	Owner
Customer master/account after conversion	ILM Customer Service
Order workflow after submission	FUL-02
Payment ledger	BIL-01-PAY-01

Out of scope	Owner
Contractor installation capacity	EM-02 + WO
Package eligibility source of truth	SIP/RLM/PLM
Final commission payroll/payout accounting	Future HR/finance or commission DD if required

Commission scope decision: SALES-01 records attribution events needed for future commission reporting, but it does not calculate payable commissions or create settlement/payroll obligations in v1. Commission payout is deliberately deferred to a future finance/commission DD so the first release can support sales attribution without introducing payroll liability logic.

3. Relationship To EM-02

EM-02 already owns contractor, staff, team, and agent registry. SALES-01 does not duplicate those records.

SALES-01 references EM agents/teams:

SALES concept	Source
Sales agent user	em_agent where agent_kind = 'sales-rep' and linked to FOUNDATION_AUTH user id.
Sales team	em_team where team_kind is sales/franchise sales.
Franchise/contractor entity	em_contractor where contractor type is FRANCHISE or configured commercial partner type.
Supervisor/team lead	em_agent_team_membership with role Lead or supervisor mapping.

Rule: do not create separate sales-user tables unless they store sales-specific state. Person/team identity remains in EM.

4. Frontend Usage

Primary frontend:

- FE-APP-02 Outdoor Sales / Franchise App

Secondary frontend:

- FE-APP-01 BSS Backoffice Web App for supervisor/admin views.

FE-APP-02 uses SALES-01 for:

- assigned leads;
- map/route lead list;
- lead capture drafts;
- duplicate lead/customer warning;
- follow-up tasks;
- lead conversion to order;
- agent/franchise performance counters.

5. Data Ownership

SALES-01 owns:

Entity	Purpose
sales_territory	Commercial coverage area/route/franchise sales zone.
sales_lead	Prospect/customer interest before a FUL order exists.
sales_lead_assignment	Current and historical assignment to agent/team/franchise.
sales_activity	Contact attempts, visit outcomes, follow-up notes.
sales_lead_package_interest	Packages/products the prospect is interested in.
sales_conversion	Link from lead to FUL order/customer/subscription after conversion.
sales_attribution_event	Immutable record used by reporting/commission logic.

SALES-01 stores references to customer/order/subscription ids after conversion, but does not own their lifecycle.

6. Core Tables

6.1 sales_territory

What it is for: defines a commercial sales area. It may align to a tech region, but it is not required to be identical. Commercial territories can be smaller, larger, or franchise-specific.

When to use it: assign leads, filter sales app maps/lists, and track sales performance by area.

Column	Type	Purpose	Sample
territory_id	UUID PK	Internal id.	terr_01J_NRB LAV_001
operator_code	TEXT	Operator scope.	WIK
territory_code	TEXT	Stable business code.	SALES-KE-NRB-LAVINGTON
name	TEXT	Display name.	Lavington Door Sales
tech_region_code	TEXT NULL	Optional ILM-CFG-02 region link.	KE-NRB-LAVINGTON
franchise_contractor_id	TEXT NULL	Optional EM contractor/franchise owner.	contractor_ke_nairobi_a
active	BOOLEAN	Whether territory can receive new leads.	true
metadata_json	JSONB	Boundary polygon, notes, route hints.	{"routeDay": "MON"}

6.2 sales_lead

What it is for: stores a prospect before they become a formal customer order.

When to use it: a sales agent meets a prospect, gets phone/name/location/interest, but the prospect has not yet submitted a full order or KYC.

Column	Type	Purpose	Sample
lead_id	UUID PK	Lead id.	lead_01J9W7Y0Z3
operator_code	TEXT	Operator scope.	WIK
lead_number	TEXT UNIQUE	Human-friendly reference.	SL-WIK-2026-000042
source_channel	TEXT	DOOR_TO_DOOR, FRANCHISE_WALKIN, CALLBACK, REFERRAL.	DOOR_TO_DOOR
status	TEXT	Lead lifecycle state.	QUALIFIED
prospect_name	TEXT	Prospect name if provided.	Jane Njeri
primary_phone	TEXT	Prospect phone.	+254712345678
homepass_id	TEXT NULL	RLM HomePass if known.	hp_01HX5...
territory_id	UUID NULL	Sales territory.	terr_01J_NRB LAV_001
geo_lat	NUMERIC NULL	Captured latitude.	-1.2831
geo_lng	NUMERIC NULL	Captured longitude.	36.7820
consent_captured	BOOLEAN	Whether prospect consent was captured.	true
duplicate_risk	TEXT	NONE, POSSIBLE_CUSTOMER, POSSIBLE_LEAD.	POSSIBLE_LEAD
created_by_agent_id	TEXT	EM agent id.	agt_01HZ_SALES_04
created_at	TIMESTAMP	Creation time.	2026-05-28T09:15:00Z
updated_at	TIMESTAMP	Last update.	2026-05-28T10:00:00Z

6.3 sales_lead_assignment

What it is for: tracks who owns the lead at any point.

When to use it: assign/reassign a lead to an agent, team, or franchise supervisor.

Column	Type	Purpose	Sample
assignment_id	UUID PK	Assignment id.	assn_01J9W8...
lead_id	UUID FK	Lead.	lead_01J9W7Y0Z3
assigned_agent_id	TEXT NULL	EM agent id.	agt_01HZ_SALES_04
assigned_team_id	TEXT NULL	EM team id.	team_sales_nrb_04
assigned_by_user_id	TEXT	Auth user id.	kc_user_sales_sup_1
reason_code	TEXT	Reason for assignment.	TERRITORY_DEFAULT
active	BOOLEAN	Current assignment marker.	true
assigned_at	TIMESTAMP	Assignment time.	2026-05-28T09:16:00Z

6.4 sales_activity

What it is for: captures visit/call/follow-up actions.

When to use it: every meaningful sales contact attempt or outcome.

Column	Type	Purpose	Sample
activity_id	UUID PK	Activity id.	act_01J9W9...
lead_id	UUID FK	Lead.	lead_01J9W7Y0Z3
agent_id	TEXT	EM agent id.	agt_01HZ_SALES_04
activity_type	TEXT	VISIT, CALL, SMS, FOLLOW_UP, CONVERSION.	VISIT
outcome_code	TEXT	Result.	INTERESTED
notes	TEXT NULL	Agent notes.	Customer wants 50 Mbps package.
next_follow_up_at	TIMESTAMP NULL	Follow-up time.	2026-05-29T08:00:00Z
created_at	TIMESTAMP	Activity time.	2026-05-28T09:20:00Z

6.5 sales_conversion

What it is for: links lead to downstream order/customer/subscription.

When to use it: after FE-APP-02 calls SALES-01 conversion endpoint and SALES-01 starts or references a FUL-02 order.

Column	Type	Purpose	Sample
conversion_id	UUID PK	Conversion id.	conv_01J9WA...
lead_id	UUID FK	Lead.	lead_01J9W7Y0Z3
order_id	TEXT	FUL-02 order id.	ord_01HZ...
customer_id	TEXT NULL	ILM customer id when created.	cust_01HX...
subscription_id	TEXT NULL	SUB-LM id when created.	sub_01HY...
converted_by_agent_id	TEXT	EM agent id.	agt_01HZ_SALES_04
converted_at	TIMESTAMP	Conversion time.	2026-05-28T10:05:00Z

7. Lead Lifecycle

Status	Meaning	Allowed next statuses
DRAFT	Captured locally or partially submitted, not ready for assignment.	NEW, CANCELLED
NEW	Lead exists and awaits assignment/qualification.	ASSIGNED, DUPLICATE, CANCELLED
ASSIGNED	Owned by agent/team.	CONTACTED, UNREACHABLE, CANCELLED
CONTACTED	Prospect reached.	QUALIFIED, NOT_INTERESTED, FOLLOW_UP
FOLLOW_UP	Future contact scheduled.	CONTACTED, QUALIFIED, NOT_INTERESTED
QUALIFIED	Prospect is eligible enough to create order.	CONVERTED, LOST, DUPLICATE
CONVERTED	FUL-02 order created/referenced.	terminal
LOST	Prospect declined or not eligible.	terminal

Status	Meaning	Allowed next statuses
DUPLICATE	Existing customer/lead/order likely exists.	terminal or reopen by supervisor
CANCELLED	Created in error or withdrawn.	terminal

8. APIs

All APIs require Authorization: Bearer <jwt>. Create/convert commands require Idempotency-Key.

API	Purpose	Caller
POST /api/sales/leads	Create lead.	FE-APP-02, Backoffice.
GET /api/sales/leads?...	List/filter leads.	FE-APP-02, Backoffice.
GET /api/sales/leads/{lead_id}	Lead detail.	FE-APP-02, Backoffice.
PATCH /api/sales/leads/{lead_id}	Update allowed lead fields.	Assigned agent/supervisor.
POST /api/sales/leads/{lead_id}/assign	Assign/reassign lead.	Supervisor/system assignment.
POST /api/sales/leads/{lead_id}/activities	Add visit/call/follow-up.	Assigned agent.
POST /api/sales/leads/{lead_id}/qualify	Run qualification checks.	Assigned agent/supervisor.
POST /api/sales/leads/{lead_id}/convert-to-order	Create FUL-02 order and mark lead converted.	Assigned agent.
GET /api/sales/territories?...	Territory list.	FE-APP-02, Backoffice.
POST /api/sales/territories	Create territory.	Sales admin.
GET /api/sales/agents/{agent_id}/daily-work	Assigned leads/tasks for mobile app.	FE-APP-02.

8.1 Create Lead Example

```
POST /api/sales/leads
Authorization: Bearer <jwt>
Idempotency-Key: sales-lead-offline-draft-8f60c
Content-Type: application/json

{
  "operatorCode": "WIK",
  "sourceChannel": "D00R_TO_D00R",
  "prospectName": "Jane Njeri",
  "primaryPhone": "+254712345678",
  "homepassId": "hp_01HX5M2K9P3Q7R4T6W8Y0Z2N4P",
  "territoryCode": "SALES-KE-NRB-LAVINGTON",
  "geo": { "lat": -1.2831, "lng": 36.7820 },
  "consentCaptured": true,
  "packageInterest": [
    { "packageId": "pkg_INET_50M_HOME", "priority": 1 }
  ]
}
```

8.2 Convert To Order Example

FE-APP-02 calls SALES-01. SALES-01 validates lead state and calls FUL-02 POST /api/orders.

```
POST /api/sales/leads/lead_01J9W7Y0Z3/convert-to-order
Authorization: Bearer <jwt>
Idempotency-Key: convert-lead_01J9W7Y0Z3
Content-Type: application/json

{
  "selectedPackageId": "pkg_INET_50M_HOME",
  "customerDraft": {
    "fullName": "Jane Njeri",
    "primaryPhone": "+254712345678",
    "idDocumentType": "NATIONAL_ID",
    "idDocumentNumber": "12345678"
  },
  "kycFileRefs": ["file_kyc_front_01", "file_kyc_back_01"],
  "salesAttribution": {
    "leadId": "lead_01J9W7Y0Z3",
    "agentId": "agt_01HZ_SALES_04",
    "teamId": "team_sales_nrb_04",
    "franchiseContractorId": "contractor_ke_nairobi_a",
  }
}
```

```

    "territoryCode": "SALES-KE-NRB-LAVINGTON"
  }
}

```

9. Integration Rules

Rule	Description
SALES-1	Lead creation must check for duplicate active lead by phone/homepass before insert. Same idempotency key returns same lead.
SALES-2	Lead conversion must be transactional inside SALES-01 for lead status/conversion row, but FUL-02 remains order owner. If FUL order creation fails, lead stays QUALIFIED with conversion failure detail.
SALES-3	SALES-01 must call ILM/RLM/SIP APIs for qualification. It must not read their DBs.
SALES-4	FE-APP-02 offline drafts are not authoritative leads until submitted to SALES-01.
SALES-5	Commission payout must not be calculated from mutable lead fields. Use immutable sales_attribution_event.
SALES-6	A converted lead cannot be reassigned unless supervisor explicitly corrects attribution through an audited endpoint.
SALES-7	If a duplicate customer/order is found, return a structured conflict and link to allowed existing record details.

10. Events

Event	Topic	Produced when	Consumers
SalesLeadCreated	sophix.sales.lead-created	Lead created.	Reporting, notifications.
SalesLeadAssigned	sophix.sales.lead-assigned	Lead assigned/reassigned.	ICN, reporting.
SalesLeadQualified	sophix.sales.lead-qualified	Qualification succeeds.	Reporting.
SalesLeadConverted	sophix.sales.lead-converted	FUL-02 order created.	Reporting, commission/payout future.
SalesLeadLost	sophix.sales.lead-lost	Lead closed lost.	Reporting.
SalesAttributionRecorded	sophix.sales.attribution-recorded	Attribution event written.	Reporting, commission/payout future.

All events use foundation outbox and include operatorCode, leadId, correlationId, eventId, schemaVersion, and occurredAt.

11. Frontend/API Mapping

FE-APP-02 screen	Module APIs
Daily work	GET /api/sales/agents/{agent_id}/daily-work, GET /api/sales/leads?...
Lead capture	POST /api/sales/leads, file upload through FOUNDATION_FILE_STORAGE.
Coverage check	GET /api/homepasses/{homepass_id}, HomePass search APIs from RLM.
Package selection	GET /api/packages/{package_id}, package list APIs from SIP.
Lead qualification	POST /api/sales/leads/{lead_id}/qualify; SALES calls ILM/RLM/SIP as needed.
Convert to order	POST /api/sales/leads/{lead_id}/convert-to-order; SALES calls FUL-02.
Follow-up	POST /api/sales/leads/{lead_id}/activities.
Supervisor lead board	SALES lead/assignment APIs, EM agent/team APIs.

12. Roles

Roles are sourced from FOUNDATION_AUTH.

Role family	Allowed actions
Sales agent	Create/update own leads, add activity, qualify, convert assigned leads.
Sales supervisor	Assign/reassign leads, view team leads, correct lead status, view performance.
Franchise manager	View franchise leads and agents, supervise assigned territories.
Sales admin	Manage territories and sales configuration.
Read-only/reporting	View only, no mutation.

13. Infra Recommendation

Budget-conscious medium deployment:

Component	Recommendation
API pods	2 pods, 1-2 vCPU, 2-4 GB RAM.
Worker/outbox	Can run in same deployment initially; split if event volume grows.
Database	Existing business DB cluster, separate sales schema.
Cache	Redis for territory/package/homepass lookup caching, not source of truth.
Storage	Use FOUNDATION_FILE_STORAGE for consent/KYC evidence.

14. Implementation Notes

Start with:

1. Create territory and lead tables.
2. Implement create/list/detail/update lead APIs.
3. Integrate EM agent/team lookup.
4. Implement duplicate detection by phone and HomePass.
5. Implement qualification wrapper calling RLM/SIP/ILM.
6. Implement convert-to-order with FUL-02.
7. Emit lead and attribution events through outbox.
8. Add FE-APP-02 screen mappings and E2E tests.

15. Acceptance Checklist

- [] FE-APP-02 does not store authoritative leads locally.
- [] Lead-to-order conversion creates a FUL-02 order.
- [] Lead attribution survives order creation.
- [] Duplicate lead/customer detection exists.
- [] EM owns sales agent/team identity.
- [] SALES events are published through outbox.
- [] E2E test covers lead capture to order submission.